

AJMAL HUSSAIN K M

Builder · Tinkerer · Coordinator

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[Portfolio](#) | [GitHub](#) | [LinkedIn](#) | [X / Twitter](#)

SUMMARY

Software developer who ships. Building agentic AI systems, hacking on IoT and embedded hardware, and running maker communities. Co-founder & Creative Lead at **Nexr**; currently interning as an IoT Assistant on the **PM Vikas** programme at IIIT Kottayam with **12+ Arduino/IoT systems shipped to GitHub** — including an AI-powered automation platform, smart-city monitor, and wearable health alert system. Serves as Membership Development Coordinator at the IEEE Student Branch, CEK.

EDUCATION & TRAINING

B.Tech (Engineering) · College of Engineering Kidangoor (CEK), Kerala 2025 – 2029

- Active IEEE Student Branch member; serving as Membership Development Coordinator.

PM Vikas IoT Programme · IIIT Kottayam May – Jun 2026

- Four-phase structured curriculum: Electronics for IoT → Networking & Cloud → Arduino Programming → Project-cum-Product Development.
- Faculty: Dr. Anand (HOD ECE), Dr. Vineeth, Dr. Della Thomas, Dr. S. Venkatesh, Shajulin Sir.

TECHNICAL SKILLS

Languages	C, C++, Python, Java
AI	Agentic AI, LLM-based agents, tool-use & reasoning, prompt engineering
IoT & Embedded	Arduino Uno (ATmega328P), GPIO, ADC, PWM, interrupts; sensors (TMP36, DHT11, HC-SR04, LDR/IR, soil/water-level); actuators (servo, DC motor, relay, buzzer); 16×2 LCD
Protocols & Networking	TCP/IP, OSI model, MQTT (pub/sub, QoS), HTTP/REST, CoAP, WebSocket; UART, SPI, I ² C
Cloud & DevOps	AWS (EC2, S3, IoT Core, Lambda, DynamoDB), Docker (images, containers, Dockerfile), Cisco Packet Tracer
Digital Electronics	Boolean algebra, K-maps, combinational & sequential logic, flip-flops, state machines, rectifiers
Tools	Arduino IDE, Git & GitHub, VS Code, Linux, virtual machines
Other	Product design, creative direction, mentoring, community building, technical writing

EXPERIENCE

Co-founder & Creative Lead · Nexr 2025 – Present

- Shape brand identity, product vision, and visual language for an early-stage venture building thoughtful digital tools.
- Lead end-to-end creative direction and product design — translating concept into shippable user experience.
- Live: nexr-one.vercel.app

IoT Assistant Intern · PM Vikas Programme — IIIT Kottayam May 2026 – Jun 2026

- Selected for the Pradhan Mantri Vishwakarma Kaushal Samman (PM Vikas) IoT internship under faculty including Dr. Anand (HOD ECE), Dr. Vineeth, Dr. Della Thomas, Dr. S. Venkatesh, and Shajulin Sir.
- Built and version-controlled **12 of 20 Arduino/IoT systems** on GitHub — energy monitoring, smart agriculture, traffic control, industrial safety, smart-city aggregator, AI-driven automation, and a wearable health alert system.
- Completed structured curriculum across Network Theory, Digital Electronics, Embedded Systems, Computer Networks, Cloud Computing (AWS, Docker), and IoT protocols (MQTT, CoAP).
- Documented **18+ daily class journals** with concepts, formulas, and exercises — published as a public learning log on the portfolio.

Membership Development Coordinator · IEEE Student Branch, College of Engineering Kidangoor 2025 – Present

- Drive membership growth and engagement at the branch — onboard new members, organize technical events, and mentor peers through first projects.
- Shipped **IEEE Volunteer Connect** — a web platform that helps IEEE volunteers find, track, and coordinate community work: ieee-vc-cek-main.web.app

Social Media Manager (Part-time) · Noon Academy

Current

- Plan content calendars, grow engagement, and maintain consistent brand voice across platforms.

Founder · Atom Innovations · Atom Skill Academy

Launching 2026

- **Atom Innovations** — building a studio for IoT, robotics, and embedded product development; from prototype to shippable hardware + software.
- **Atom Skill Academy** — preparing an EdTech space to teach students programming, electronics, and project-based engineering skills.

FEATURED PROJECTS

AI-Powered Smart Automation & Monitoring Platform ★ Arduino Uno · C++ · AI [View repo ↗](#)

- Flagship build fusing AI logic with embedded sensing — the system makes autonomous real-time decisions rather than just reacting to thresholds. Ships with full code and a detailed project report.

IoT Smart City Monitoring & Management System Arduino · C++ · IoT [View repo ↗](#)

- Centralized IoT platform aggregating sensor data across urban infrastructure — a building block for connected smart cities.

Smart Health Monitoring & Emergency Alert System Arduino · C++ · Wearables [View repo ↗](#)

- Wearable-style vitals tracker that fires an automatic SOS alert when readings cross safe thresholds — so help arrives faster.

IEEE Volunteer Connect Web App · Firebase [View repo ↗](#)

- Live community platform for IEEE volunteers to find, track, and coordinate work — replaces ad-hoc spreadsheets with a single source of truth.

Smart Energy Monitoring & Management System Arduino Uno · Power Electronics [View repo ↗](#)

- Real-time electrical-parameter measurement to track and optimize household power usage.

Temperature-Controlled Fan with LCD Readout TMP36 · DC Fan · 16×2 LCD [View repo ↗](#)

- Closed-loop thermal management — analog temperature read, automatic fan switching, live LCD display.

Six additional builds shipped to GitHub: smart home security, smart parking gate, smart agriculture, water-level + pump control, smart weather station, smart traffic-light controller, industrial safety monitor — all linked from [portfolio-8a1w.vercel.app/pm-vikas.html](https://8a1w.vercel.app/pm-vikas.html).